

Introduction to Demographic Methods

(4: Population projections)

Aliakbar Akbaritabar^{1,2}

¹Max Planck Institute for Demographic Research, Rostock, Germany;

²Institute of Sociology and Demography, University of Rostock, Rostock, Germany;

akbaritabar@demogr.mpg.de; aliakbar.akbaritabar@uni-rostock.de

Introduction to Demographic Methods



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- 1- Introduction. Why study Demography? ... Git, version control, course materials GitHub
- 2- Mortality
- 3- Fertility
- 4- Population projections
- 5- Migration
- 6- Kinship
- 7- The future, and Digital and Computational Demography
- 9- Wrap up + Description of examination
- 10- Final examination: 5 min presentation to share your selected topic + 5 min Q&A; 6 weeks to prepare a short essay and empirical analysis

Discussing previous week's materials

1) Slides, Readings

1. Any questions on my slides?
2. Let's discuss the "key reading material".
3. Example: Discuss at least 1-2 points, e.g., 1) What was surprising and counterintuitive for you? 2) What did you find that was already expected? 3) What could be extended in this reading/work?
4. Any thoughts about the additional reading materials?

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2) Hands-on

1. Any problems in installing the requirements?
2. Any issues in running the codes in hands-on materials to reproduce the examples?
3. Did anyone extend the examples? How? What did you do and find?

Few points (TBA)



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Switch to Ernesto Amaral's slides: "Lecture05" and Monica
Alexander's slides: "5_pop_proj"

Hands-on part

Needed installations (or checking requirements)

For today's hands-on part, see the Readme file under "03_MonicaALabs" folder and download the requested files under "data" folder.

We will go over the notebook called "5_pop_proj_edits.qmd" and see what it does. You need to activate Renv and install R packages listed in the first code block of the notebook, i.e., "tidyverse, here, readxl, janitor" (please check the notebook for the latest list).

Needed: R, RStudio, and Renv environment.

Where to learn basics of R and Python?

To start with Python

Check this website first:

https://www.w3schools.com/python/python_intro.asp

Check this repository by Vincent Traag and others, for an introductory course and code:

<https://github.com/vtraag/intro-python>

Or this one by Data Carpentry:

<https://datacarpentry.org/python-ecology-lesson/>

To start with R

Check this website first:

<https://www.w3schools.com/r/default.asp>

This “very short introduction to R” is a good start:

<https://cran.r-project.org/doc/contrib/Torfs+Brauer-Short-R-Intro.pdf>

Or this course by Data Carpentry:

<https://datacarpentry.org/R-genomics/index.html>

Reading materials for today



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Key reading material, [Chapters 5 and 10](#)

 [Wachter, K. W. \(2014, June\). Essential Demographic Methods. Harvard University Press.](#)

Additional reading materials

 [Alexander, M. \(2026\). Advanced Data Analysis course in Demographic Methods. Retrieved April 8, 2026, from <https://github.com/MJAlexander/soc6708>](#)

 [Preston, S., Heuveline, P., & Guillot, M. \(2000\). Demography: Measuring and Modeling Population Processes. Wiley.](#)

What to do before the next session?

1) Slides, Readings

1. Study my slides carefully.
2. Check the slide with reading materials.
3. Read the “key reading material”.
4. Prepare at least 1-2 discussion points, e.g., What was surprising and counterintuitive for you? What did you find that was already expected? What could be extended in this reading/work?
5. Skim over the additional materials.

2) Hands-on

1. Check the slide and folder for hands-on materials.
2. Install the requirements as outlined.
3. Run the codes in hands-on materials to reproduce the examples, and if you can extend them.
4. Note down the unclear points, errors, and questions for the next session.

Thanks for your attention!

Questions and comments are always welcome!

Aliakbar Akbaritabar

Assistant Professor of Computational Social Science
University of Rostock, and Max Planck Institute for
Demographic Research (MPIDR), Rostock, Germany.

To contact, please use Email:

“aliakbar.akbaritabar@uni-rostock.de” and include in the
subject line: “[Demogr-SoSe-2026] your subject”

LinkedIn/BlueSky/Twitter/Mastodon: @akbaritabar

References



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